

Tutorial 10

CS105.3 Database Management Systems

ER Diagrams, Relational Schema Mapping and Normalization

Question 01

EMC (Pvt) Ltd has decided to appoint you as the database specialist to design a database to store information about musicians who perform on their albums (as well as other company data). The basic information available is as follows.

- Each musician that records at EMC (Pvt) Ltd has an ID, a name, an address and a phone number. Some musicians prefer to share the same address, and no address has more than one phone.
- Each instrument that is used in songs recorded at EMC (Pvt) Ltd has a name (e.g. violin, guitar) and a musical key (e.g., C, D-flat, E-flat).
- Each album that is recorded at the EMC (Pvt) Ltd label has a title, a copyright date, a format (e.g., CD) and an album identifier.
- Each song recorded in EMC (Pvt) Ltd has a song ID, title and an author.
- Each musician may play several instruments, and a given instrument may be played by several musicians.
- Each album has several songs on it but no song may appear on more than one album.
- Each song is performed by one or more musicians, and a musician may perform several songs.
- Each album has exactly one musician who acts as its producer. A producer may produce several albums.

- (i) Draw an ER diagram and map the relational schema for this scenario. Discuss any assumptions you make.

Question 02

Normalize below to 3NF.

Make sure that you show the intermediate steps (i.e., 1NF, 2NF and 3NF). Designate the Primary Keys (underline) and Foreign Keys (dotted underline) for each table.

MusicianID	MusicianName	Address	Phone	Instrument Name	Instrument Key	AlbumID	Album Title	Copyright Date	Format	Song ID	Song Title	Song Author	ProducerID
M001	John Doe	123 Elm St, City	555-1234	Guitar	C	A001	Album1	2022-01-01	CD	S001	Song1	Author1	P001
M002	Jane Smith	456 Oak St, City	555-5678	Violin	D-flat	A001	Album1	2022-01-01	CD	S001	Song1	Author1	P001
M001	John Doe	123 Elm St, City	555-1234	Guitar	C	A001	Album1	2022-01-01	CD	S002	Song2	Author2	P001
M002	Jane Smith	456 Oak St, City	555-5678	Violin	D-flat	A002	Album2	2023-01-01	CD	S003	Song3	Author3	P002
M003	Jim Brown	789 Pine St, City	555-7890	Piano	E-flat	A002	Album2	2023-01-01	CD	S003	Song3	Author3	P002